FRAUD RISK INTELLIGENCE: AN EXPLAINABLE AND COST-AWARE FRAUD DETECTION FRAMEWORK

GAMAGE RECRUITERS (PVT) LTD

Prepared by: T.M.T.S.B. Tennakoon

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FRAUD RISK INTELLIGENCE

An Explainable and Cost-Aware Fraud Detection Framework

Final Business Fraud Strategy Report

Project: Fraud Risk Intelligence
Report Type: Business Strategy and Model Evaluation
Implementation: Python-Based Machine Learning Framework
Evaluation Dataset: 3,000 Financial Transactions
Final Deployment Recommendation: Research-Only — Not Approved for Live Automated Blocking

1. EXECUTIVE SUMMARY

Financial transaction fraud represents a significant challenge for financial institutions, payment service providers, and digital banking platforms. Fraudulent activities can result in financial losses, customer dissatisfaction, operational disruptions, and reputational damage.

Traditional fraud detection approaches frequently depend on static rules and predefined transaction thresholds. Although these approaches are relatively easy to implement, they may struggle to identify complex fraudulent behavior and adapt to changing transaction patterns.

This project developed an explainable, machine learning-based Fraud Risk Intelligence Framework designed to identify potentially fraudulent financial transactions and support risk-based operational decision-making.

The proposed framework integrates data quality assessment, exploratory data analysis, feature engineering, machine learning, class imbalance handling, probability calibration, cost-sensitive decision optimization, fraud operations simulation, explainable artificial intelligence, robustness testing, and model governance.

The project analyzed a dataset containing 3,000 financial transactions, including 364 fraudulent transactions and 2,636 legitimate transactions. The overall fraud prevalence was approximately 12.13%.

Several machine learning approaches were evaluated, including Logistic Regression, Random Forest, and XGBoost. Additional experiments investigated class weighting, random oversampling, and probability calibration.

Following model development and validation, an Unweighted Logistic Regression model with Sigmoid Probability Calibration was selected as the provisional scoring model.

The model was combined with a three-level fraud decision strategy consisting of transaction approval, manual review, and proposed transaction blocking.

The final evaluation was conducted using a reserved holdout dataset of 600 transactions, containing 73 fraudulent cases.

The final model achieved an Average Precision score of 0.1932, an ROC-AUC of 0.6640, and a Brier Score of 0.1039.

The proposed decision policy approved 474 transactions, assigned 114 transactions for manual review, and proposed blocking 12 transactions.

Under hypothetical business cost assumptions, the policy produced an estimated scenario cost of LKR 997,830 compared with LKR 1,460,000 for an approve-all strategy. This represents an illustrative cost reduction of approximately 31.66%.

However, the final evaluation also identified significant limitations. The policy approved 46 of the 73 fraudulent transactions, while eight of the twelve proposed blocks involved legitimate transactions.

Consequently, although the framework demonstrates the potential value of combining machine learning predictions with cost-sensitive decision-making, its current performance is insufficient for unsupervised production deployment.

The principal recommendation is to retain the framework as a research prototype and decision-support system until additional validation, operational safeguards, and governance requirements are satisfied.

2. INTRODUCTION

2.1 Background

The increasing adoption of digital payments, mobile banking, e-commerce, and online financial services has created new opportunities for fraudulent transactions.

Financial institutions must continuously evaluate transaction behavior to detect suspicious activities while minimizing unnecessary disruption to legitimate customers.

Fraud detection is particularly challenging because fraudulent transactions often represent a relatively small proportion of all transactions.

This creates a class imbalance problem in which machine learning models may favor legitimate transactions and fail to detect important fraudulent cases.

Furthermore, incorrectly identifying legitimate transactions as fraudulent can negatively affect customer trust and increase operational expenses.

Therefore, an effective fraud detection framework must consider not only predictive accuracy but also fraud detection performance, probability reliability, investigation workload, customer impact, and financial consequences.

2.2 Business Problem

The central business problem addressed by this project is:

How can machine learning be used to identify potentially fraudulent financial transactions while balancing fraud prevention, operational capacity, customer inconvenience, and financial costs?

A conventional binary classification system produces a fraud or legitimate prediction.

However, financial institutions typically require more flexible decisions.

For example, transactions with relatively low predicted fraud risk may be approved, transactions with moderate risk may require additional verification, and transactions with higher risk may be considered for blocking.

The challenge is determining suitable decision thresholds without creating excessive manual reviews or unnecessary customer disruption.

2.3 Project Objectives

The primary objective of this project was to develop and evaluate a machine learning-based fraud risk intelligence framework that supports explainable and cost-aware transaction decisions.

The specific objectives were to:

1. Investigate transaction data quality, missing values, outliers, and potential data leakage.
2. Identify fraud-related patterns through exploratory data analysis.
3. Develop relevant engineered features from transaction behavior.
4. Compare baseline and advanced machine learning models.
5. Evaluate techniques for handling class imbalance.
6. Improve the reliability of fraud probability estimates through calibration.
7. Develop a cost-sensitive transaction decision policy.
8. Simulate fraud investigation operations and manual review workloads.
9. Explain model predictions using SHAP.
10. Evaluate model stability under data perturbations.
11. Establish governance controls and assess deployment readiness.
12. Evaluate the frozen framework using a final holdout dataset.

3. DATASET DESCRIPTION AND DATA QUALITY

3.1 Dataset Overview

The dataset contained 3,000 financial transaction records with 25 original columns.

Each row represented an individual financial transaction.

The dataset included transaction characteristics, customer information, transaction behavior, device information, authentication details, and risk-related indicators.

The target variable, fraud, represented whether a transaction was fraudulent.

A value of 1 indicated a fraudulent transaction, while a value of 0 represented a legitimate transaction.

Table 3.1Dataset Summary

Description	**Value**
Total Transactions	3,000
Original Columns	25
Legitimate Transactions	2,636
Fraudulent Transactions	364
Fraud Percentage	12.13%
Legitimate Percentage	87.87%
Total Missing Values	582
Transactions with Missing Values	542
Duplicate Transactions	0
Duplicate Transaction IDs	0

The dataset demonstrated a moderate class imbalance, with legitimate transactions accounting for approximately 87.87% of the observations.

Consequently, overall classification accuracy alone would not provide a reliable measure of fraud detection performance.

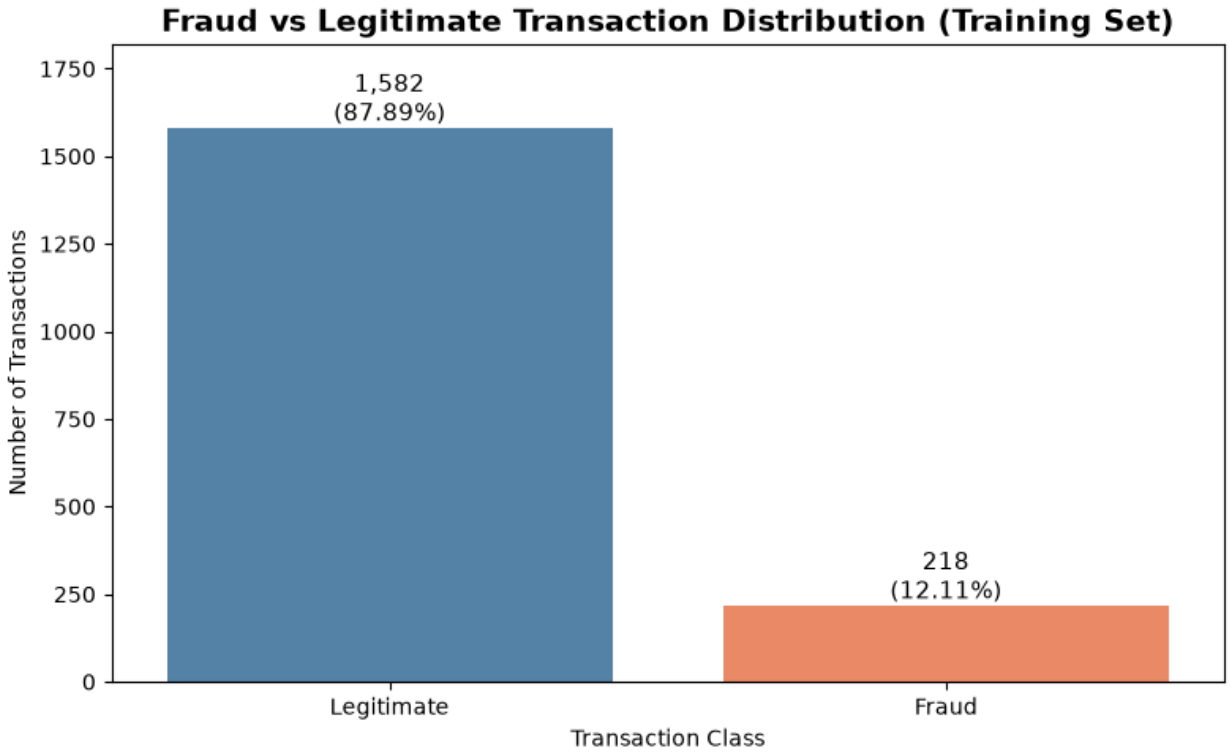


Figure 3.1 Fraud vs. Legitimate Transaction Distribution in the Training Dataset.

3.2 Data Quality Assessment

A structured data quality investigation was conducted to identify missing information, invalid values, duplicate records, unusual observations, and possible data leakage.

The investigation identified 582 missing values across the dataset.

The authentication_level feature contained the largest number of missing observations, with 289 missing records.

Other features containing missing values included transaction amount, transaction distance, merchant risk score, IP risk score, and login-to-transaction duration.

Missing authentication information was particularly important because exploratory analysis indicated that transactions with missing authentication levels exhibited elevated observed fraud rates.

Therefore, missing values were not automatically removed from the dataset.

Instead, appropriate imputation methods and missingness indicators were considered during feature engineering and model preprocessing.

3.3 Outlier Investigation

An Interquartile Range (IQR) analysis was performed to identify unusually high or low numerical observations.

The analysis identified 565 unique transactions containing at least one IQR outlier.

These represented approximately 18.83% of the dataset.

Transactions containing outliers exhibited an observed fraud rate of 18.41%, compared with 10.68% for transactions without identified outliers.

This suggested that unusual transaction characteristics could contain useful fraud detection information.

Consequently, numerical outliers were retained rather than automatically removed.

3.4 Dataset Partitioning

The dataset was divided into training, validation, and testing subsets using stratified sampling.

Table 3.2 Dataset Partitioning

Dataset	Transactions	Fraud Cases	Fraud Rate
Training	1,800	218	12.11%
Validation	600	73	12.17%
Testing	600	73	12.17%

The training dataset was used for model fitting and cross-validation.

The validation dataset supported model comparison, calibration assessment, and decision policy selection.

The testing dataset was reserved for the final evaluation after the model and decision thresholds were frozen.

Preprocessing and resampling operations were incorporated into training pipelines to reduce leakage between cross-validation folds.

Nevertheless, the absence of complete timestamps and customer identifiers prevented strict chronological and customer-level validation.

Additionally, preliminary data exploration had examined the complete dataset during the early stages of development.

Therefore, the final test results should be interpreted as an internally held-out evaluation rather than fully independent external validation.

4. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was conducted to investigate relationships between transaction characteristics and fraudulent activity.

The analysis examined categorical variables, numerical distributions, transaction behavior, risk scores, and interactions between multiple features.

4.1 Fraud Distribution Across Transaction Channels

Fraud rates varied across transaction channels.

E-commerce transactions exhibited the highest observed fraud rate at approximately 15.00%.

Mobile application transactions exhibited a fraud rate of 11.76%, followed by ATM transactions at 9.85% and POS transactions at 9.54%.

These results suggested that transaction channels could provide useful contextual information for fraud detection.

However, the observed differences did not establish that specific transaction channels directly caused fraudulent activity.

4.2 Transaction Velocity

Transaction velocity represented the number of transactions occurring within a specified time window.

Analysis of transactions occurring during the previous hour revealed an association between higher transaction frequency and increased fraud risk.

Transactions with seven or more recorded activities within an hour exhibited a fraud rate of 45.45%.

However, this group contained only eleven transactions.

The relatively small sample size meant that the observed rate was subject to considerable uncertainty.

A larger group containing transactions with three to four recent activities exhibited a fraud rate of 15.34%.

These findings suggested that transaction velocity could contribute valuable predictive information.

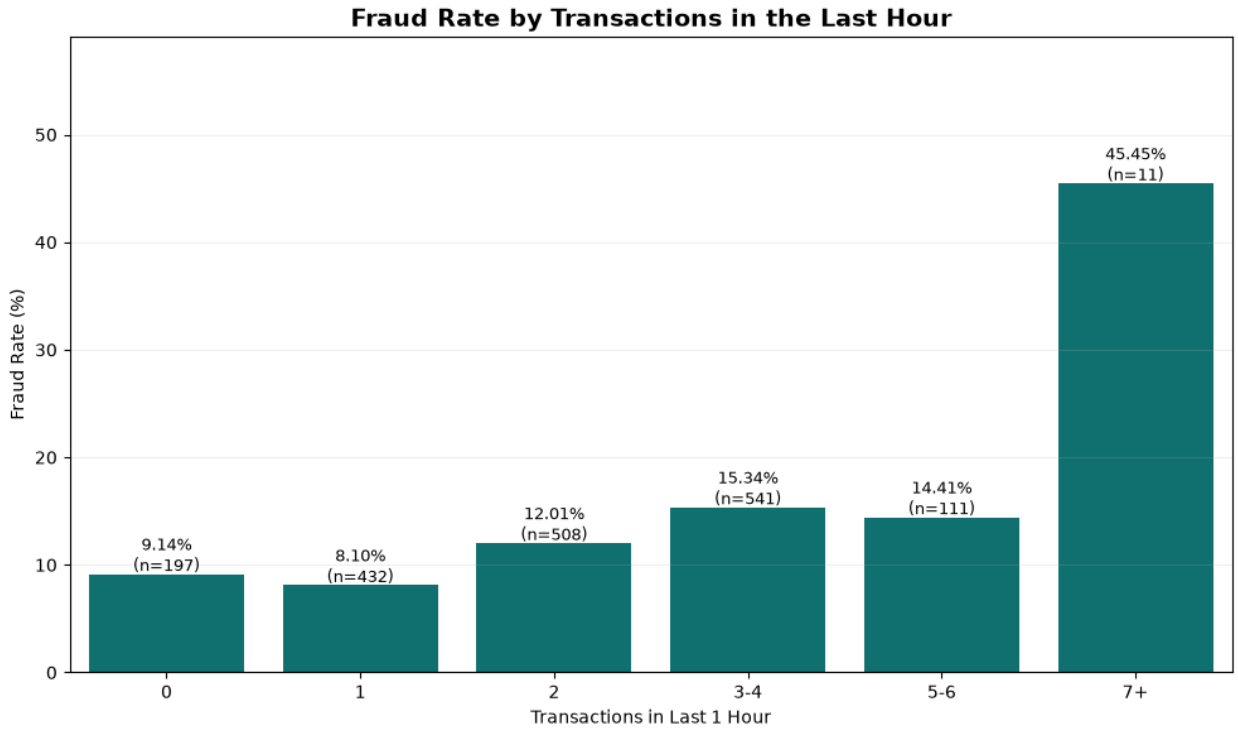


Figure 4.1 Fraud Rate by Transaction Velocity Within the Previous Hour.

4.3 Transaction Amount

The analysis investigated fraud rates across five transaction amount quantiles.

The highest transaction amount quantile exhibited a fraud rate of 16.95%.

However, fraud rates did not increase consistently across all amount groups.

This indicated that the relationship between transaction amount and fraud was not strictly monotonic.

As a result, nonlinear transformations and interactions involving transaction amount were considered during feature engineering.

4.4 Customer Account Age

Older customer accounts generally exhibited lower observed fraud rates.

The oldest account age group recorded a fraud rate of 9.17%, while the youngest account age group recorded 13.85%.

However, intermediate groups displayed variations, indicating that the relationship was not consistently linear.

Account age was therefore retained as a potentially useful predictive feature.

4.5 Feature Interaction Analysis

Several interaction analyses were conducted to investigate whether combinations of transaction characteristics provided additional information.

Transactions involving a new device and a distance exceeding approximately 49 kilometers from the usual location exhibited a fraud rate of 23.17%.

Transactions involving high transaction amounts and high merchant risk scores exhibited a fraud rate of 18.28%.

Similarly, E-commerce transactions with missing authentication information exhibited a fraud rate of 27.12%.

These findings suggested that combinations of features could contain useful predictive relationships.

However, the observed differences were exploratory associations rather than confirmed causal or statistical interaction effects.

5. FEATURE ENGINEERING

Feature engineering was performed to transform original transaction characteristics into potentially more informative predictive variables.

The original modeling dataset contained 23 candidate input features after excluding the transaction identifier and fraud target.

Sixteen additional features were developed, increasing the total number of candidate predictors to 39.

Table 5.1 Engineered Feature Categories

Feature Category	**Examples**
Transaction Transformations	amount_log1p, distance_log1p
Transaction Velocity	velocity_ratio_1h_24h, hourly_velocity_intensity
Behavioral Indicators	high_velocity_indicator, night_transaction_indicator
Transaction Normalization	amount_per_recent_transaction
Risk Combinations	combined_risk_score, risk_score_product
Risk Interactions	amount_merchant_risk_interaction
Device Behavior	new_device_distance_interaction
Customer Behavior	device_account_age_ratio
Authentication	authentication_missing_indicator
Context Interactions	authentication_channel_interaction

Feature engineering was implemented through a reusable Python module.

The module was designed to preserve original input records and avoid directly using the fraud target to construct derived predictors.

A comparison was conducted between models trained using original and engineered feature sets.

The engineered Logistic Regression model achieved a cross-validation Average Precision of 0.2457, compared with 0.2362 for the original-feature model.

However, the engineered model achieved a lower validation Average Precision of 0.2203 compared with 0.2424 for the original-feature model.

These results demonstrated that adding engineered features did not necessarily improve generalization.

Therefore, engineered features were treated as experimental predictors whose usefulness required empirical validation.

6. MACHINE LEARNING MODEL DEVELOPMENT

6.1 Model Selection

Three machine learning algorithms were investigated:

Logistic Regression: A statistical classification method that provides an interpretable baseline.

Random Forest: An ensemble learning algorithm capable of capturing nonlinear relationships.

XGBoost: A gradient-boosting algorithm designed to learn complex predictive patterns.

Model development used training-only cross-validation and a separate validation dataset.

Random Forest and XGBoost were evaluated using limited hyperparameter searches.

6.2 Validation Performance

Table 6.1 Machine Learning Model Comparison

Model	Validation PR-AUC	ROC-AUC
Logistic Regression — Original	0.2424	0.6895
Logistic Regression — Engineered	0.2203	0.6821
Random Forest — Engineered	0.2249	0.6474
XGBoost — Engineered	0.2130	0.6837

The original-feature Logistic Regression configuration achieved the highest validation PR-AUC and ROC-AUC among these initial model development experiments.

The advanced models did not consistently outperform the simpler baseline.

This demonstrated the importance of empirical evaluation rather than assuming that more complex algorithms would automatically provide superior results.

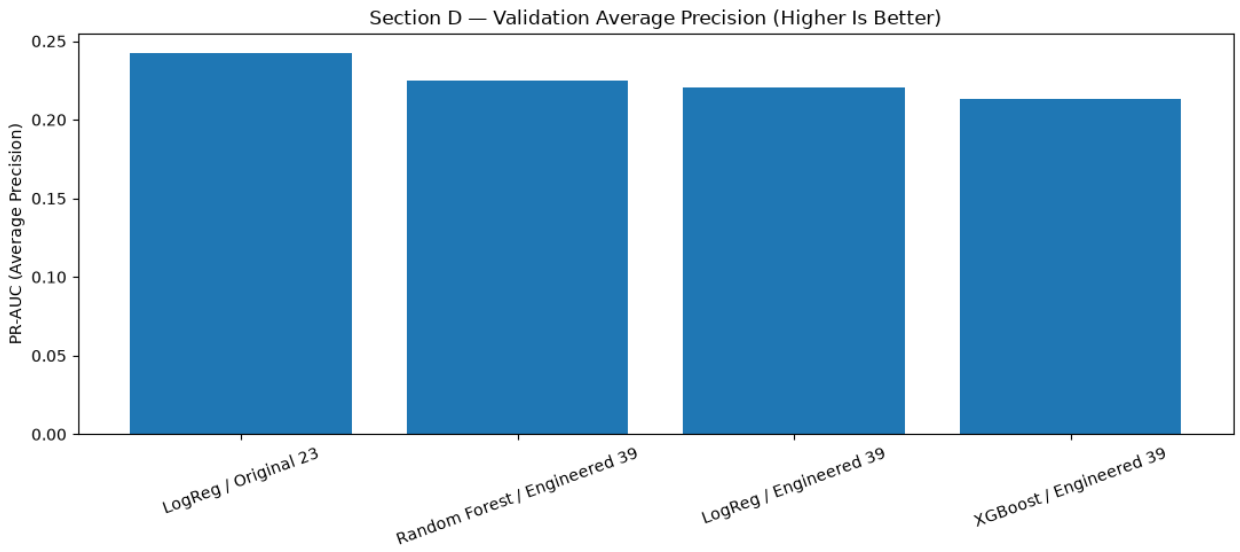


Figure 6.1 Validation Average Precision Comparison Across Machine Learning Models.

6.3 Handling Class Imbalance

Three imbalance-handling strategies were evaluated:

- Unweighted Logistic Regression.
- Class-weighted Logistic Regression.
- Random Oversampling.

At the default classification threshold of 0.50, the class-weighted and oversampling approaches achieved substantially higher fraud recall than the unweighted model.

However, these approaches also generated considerably more false positives.

The unweighted model identified only three of the 73 fraudulent validation transactions at this threshold.

The class-weighted model identified 46 fraudulent transactions, while random oversampling identified 47.

However, class weighting generated 187 false positives, while random oversampling generated 180.

These findings highlighted the trade-off between identifying fraudulent transactions and unnecessarily flagging legitimate transactions.

Consequently, the project did not rely solely on the default classification threshold.

Instead, calibrated probabilities were used to support a subsequent cost-sensitive decision strategy.

7. PROBABILITY CALIBRATION

Probability calibration was applied to improve the reliability of predicted fraud probabilities.

A well-calibrated model should produce predicted probabilities that correspond reasonably closely to observed outcome frequencies.

The project evaluated three calibration configurations:

1. Uncalibrated probabilities.
2. Sigmoid calibration.
3. Isotonic regression.

Calibration performance was assessed using Brier Score, Log Loss, Expected Calibration Error, PR-AUC, and ROC-AUC.

Table 7.1 Selected Calibration Results

Model Configuration	**PR-AUC**	**Brier Score**	**ECE**
Unweighted + Sigmoid	0.2408	0.1016	0.0155
Unweighted + Isotonic	0.2438	0.1017	0.0194
Class-weighted + Sigmoid	0.2426	0.1025	0.0102
Oversampling + Isotonic	0.2477	0.1017	0.0099

The Unweighted Logistic Regression model with Sigmoid Calibration achieved the lowest validation Brier Score among the evaluated configurations.

It was therefore selected as the provisional model for subsequent cost-sensitive decision experiments.

However, this selection did not establish that the configuration was universally superior across all business objectives.

Additional independent evaluation would be required to establish whether another model could produce lower operational costs.

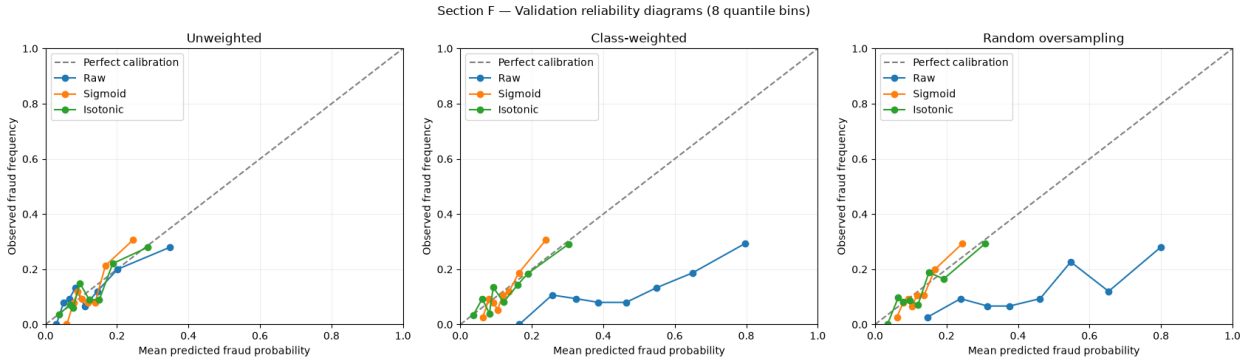


Figure 7.1 Validation Reliability Diagrams Before and After Probability Calibration.

8. COST-SENSITIVE FRAUD DECISION STRATEGY

8.1 Business Cost Framework

Fraud detection decisions involve different types of financial and operational consequences.

Approving fraudulent transactions may result in direct financial losses.

Blocking legitimate transactions can inconvenience customers and negatively affect customer trust.

Manual reviews require operational resources and may introduce transaction delays.

A hypothetical business cost framework was therefore developed to evaluate alternative decision strategies.

Table 8.1 Assumed Business Costs

Business Event	**Assumed Cost**
Fraudulent Transaction Approved	LKR 20,000
Legitimate Transaction Blocked	LKR 1,500
Manual Transaction Review	LKR 150
Manual Review Fraud Detection Rate	90%
Manual Review False Rejection Rate	2%
Maximum Review Capacity	20%
Maximum Proposed Block Rate	5%

These assumptions were introduced for research and simulation purposes.

They were not obtained from verified institutional financial records or actual fraud investigation outcomes.

8.2 Decision Threshold Optimization

A three-level transaction decision policy was developed.

Table 8.2 Optimized Decision Policy

Fraud Probability	Proposed Decision
Below 0.152046	APPROVE
0.152046 to below 0.262003	MANUAL REVIEW
0.262003 and above	PROPOSE BLOCK

The thresholds were selected using validation-stage information and hypothetical operational costs.

The optimization also considered manual review capacity and proposed block-rate restrictions.

Once selected, the model and decision thresholds were frozen before final testing.

8.3 Validation-Stage Cost Comparison

Table 8.3 Validation Business Cost Comparison

Policy	Projected Cost (LKR)
Approve All	1,501,171
Binary Blocking at 0.50	1,501,171
Optimized Three-Level Policy	990,944

The optimized policy produced an estimated reduction of approximately 34% in projected validation costs compared with the approve-all strategy.

However, this result depended on calibrated probability estimates and hypothetical cost assumptions.

The project also conducted sensitivity analysis to evaluate how changing fraud loss, false-blocking cost, and reviewer effectiveness affected the preferred decision thresholds.

The analysis demonstrated that higher legitimate-blocking costs could make blocking less economically attractive.

Therefore, the optimal policy was dependent on the assumed business environment.

9. FRAUD OPERATIONS SIMULATION

An operational simulation was developed to demonstrate how calibrated fraud probabilities could support transaction routing and investigation activities.

The simulation assigned each transaction to one of three actions: APPROVE, REVIEW, or PROPOSE BLOCK.

For transactions requiring manual review, a simulated investigation queue was generated.

The simulation assumed four fraud analysts, each with a review capacity of 30 transactions per batch.

During validation, 120 transactions were allocated for manual review.

Each analyst received 30 cases, producing a theoretical backlog of zero.

However, the simulated workload represented assumed operational capacity rather than measured employee productivity or service-level performance.

9.1 Final Test Operational Distribution

The frozen decision policy was subsequently evaluated on the final testing dataset.

Table 9.1 Final Transaction Decision Distribution

Decision	Transactions	Percentage
Approved	474	79%
Manual Review	114	19%
Proposed Block	12	2%
Total	600	100%

The final testing batch remained within the assumed operational limits.

The review rate was 19%, compared with the maximum allowance of 20%.

The proposed block rate was 2%, compared with the maximum allowance of 5%.

However, the framework should not be considered operationally reliable solely because these constraints were satisfied in one batch.

Stress testing demonstrated that changes in transaction behavior could produce capacity violations.

10. FINAL HOLDOUT MODEL EVALUATION

10.1 Evaluation Methodology

The final evaluation used the reserved testing dataset containing 600 transactions.

The frozen model, probability calibration configuration, and decision thresholds were applied without retraining or post-test optimization.

The final testing dataset contained 73 fraudulent transactions and 527 legitimate transactions.

10.2 Predictive Performance

Table 10.1 Validation and Final Test Performance

Metric	Validation	Final Test
Average Precision	0.2408	0.1932
ROC-AUC	0.6928	0.6640
Brier Score	0.1016	0.1039
Expected Calibration Error	0.0155	0.0215
Fraud Prevalence	12.17%	12.17%

The final Average Precision score decreased from 0.2408 to 0.1932.

This represented a relative reduction of approximately 19.8%.

The ROC-AUC also decreased from 0.6928 to 0.6640.

These differences suggested that the model's predictive performance was less favorable on the final holdout dataset than on the validation dataset.

Nevertheless, the final Average Precision remained above the fraud prevalence reference value of approximately 0.1217.

This indicated that the model retained some ability to rank fraudulent transactions above legitimate transactions.

However, the performance was not sufficient to establish production reliability.

10.3 Final Fraud Detection Outcomes

Table 10.2 Final Fraud Routing Results

Fraud Outcome	Transactions
Fraudulent Transactions Approved	46
Fraudulent Transactions Sent for Review	23
Fraudulent Transactions Proposed for Block	4
Total Fraudulent Transactions	73

The final decision policy approved 46 of the 73 known fraudulent transactions.

This represented approximately 63.01% of fraudulent cases.

Only 27 fraudulent transactions were routed away from automatic approval.

Furthermore, eight of the twelve proposed blocks involved legitimate transactions.

This meant that approximately 66.67% of the proposed blocking decisions were associated with legitimate customers.

These findings demonstrated two significant operational challenges.

First, a substantial proportion of fraudulent transactions remained undetected by the triage policy.

Second, the proposed blocking mechanism generated a considerable proportion of false blocking recommendations.

Consequently, fully automated transaction blocking could introduce unacceptable customer impact.

11. FINAL BUSINESS COST EVALUATION

The frozen decision policy was evaluated using the final testing dataset and the previously defined hypothetical cost assumptions.

Table 11.1 Final Business Cost Comparison

Metric	Amount (LKR)
Approve-All Scenario Cost	1,460,000
Proposed Policy Scenario Cost	997,830
Illustrative Cost Difference	462,170
Illustrative Cost Reduction	31.66%
Projected Policy Cost	998,406

The outcome-based simulation estimated a total cost of LKR 997,830 for the proposed decision strategy.

This was lower than the LKR 1,460,000 estimated for approving all transactions.

The illustrative difference was LKR 462,170, equivalent to approximately 31.66%.

However, the estimated reduction should not be interpreted as verified savings.

The result depended on assumed fraud losses, manual review expenses, customer blocking costs, and reviewer effectiveness.

The simulation also estimated that 24.7 fraudulent transactions could be prevented through proposed blocking and manual review, assuming a 90% reviewer fraud detection rate.

This was a model-based expected value rather than an observed number of prevented fraudulent transactions.

The cost analysis therefore demonstrated the potential financial value of risk-based decision-making under the selected assumptions.

Additional institutional evidence would be required before these findings could support actual operational or financial decisions.

12. EXPLAINABLE ARTIFICIAL INTELLIGENCE

Explainable Artificial Intelligence was incorporated into the framework to improve transparency and support the interpretation of fraud risk predictions.

The project used SHAP to investigate both global feature importance and individual transaction explanations.

12.1 Global Feature Importance

Global SHAP analysis identified the features that contributed most strongly to variation in the underlying Logistic Regression model scores.

Table 12.1 Top Global SHAP Features

Rank	Feature	Mean Absolute SHAP
1	Merchant Category	0.31824
2	Transactions in Last Hour	0.26547
3	New Device	0.25423
4	Transaction Channel	0.20099
5	Merchant Risk Score	0.17599

Merchant category, transaction velocity, and new-device indicators were among the most influential features.

These findings were broadly consistent with patterns observed during exploratory data analysis.

However, feature importance did not establish causation.

The reported SHAP values represented contributions to the underlying model's log-odds rather than direct changes in calibrated fraud probabilities.

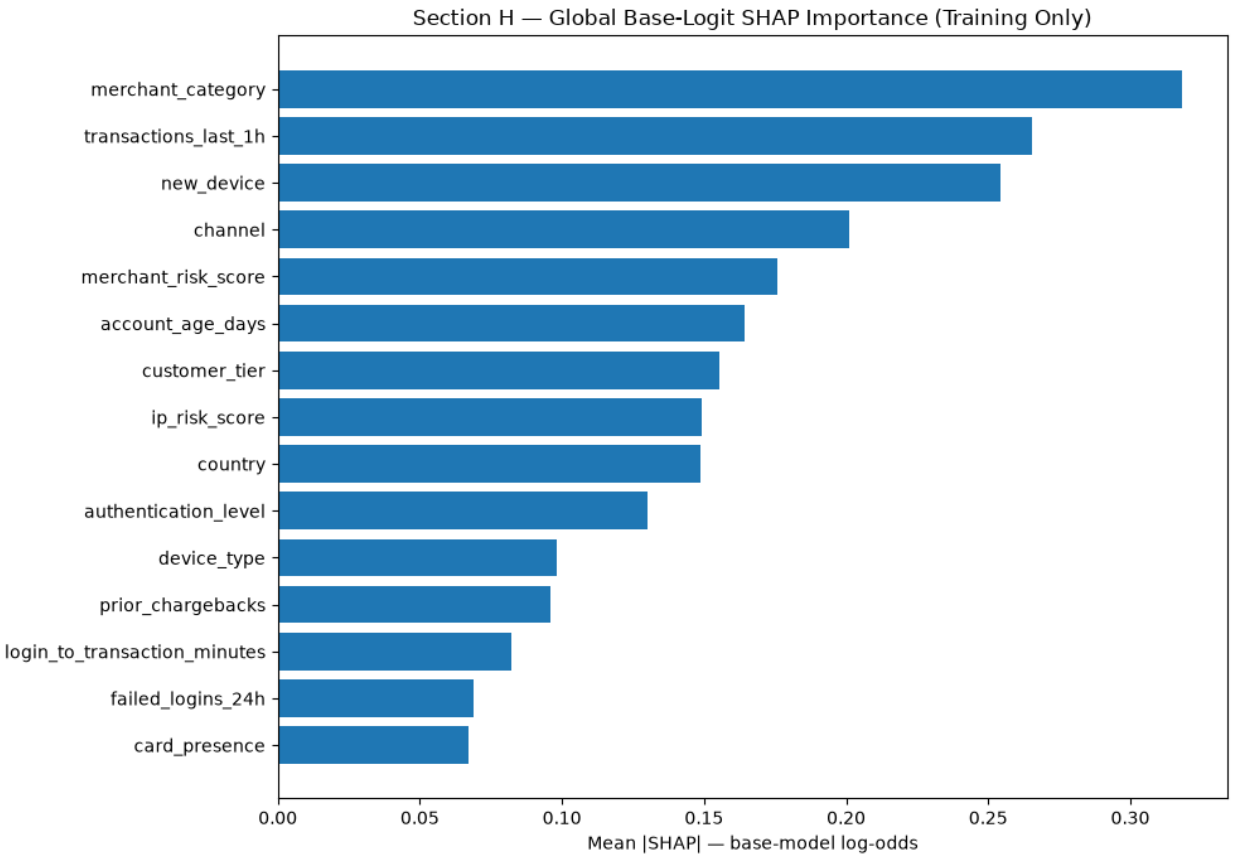


Figure 12.1 Global SHAP Feature Importance for the Underlying Logistic Regression Model.

12.2 Local Transaction Explanation

An individual transaction explanation was generated for transaction TXN5000873.

The transaction received a calibrated fraud probability of approximately 44.17%.

Under the frozen policy, the system proposed a BLOCK action.

The main upward contributions to the underlying model score included:

- High transaction amount.
- New-device indicator.
- Merchant risk score.

Conversely, merchant category and IP risk score contributed downward effects relative to the model's explanation baseline.

The analysis demonstrated how explainability methods could provide contextual information to fraud investigation teams.

However, SHAP explanations should be used as supporting evidence rather than definitive explanations of fraudulent intent.

They should not independently justify customer-impacting automated decisions.

13. MODEL ROBUSTNESS AND STRESS TESTING

Robustness testing was conducted to investigate how the frozen model and decision policy responded to changes in transaction data.

The experiments simulated missing values, altered authentication information, unseen transaction channels, increased transaction amounts, greater transaction distances, increased transaction velocity, and noisy risk scores.

The model was not retrained during these tests.

The original decision thresholds were retained to evaluate policy stability under changing inputs.

13.1 Stress Testing Results

Table 13.1 Selected Stress Testing Results

Scenario	Review Rate	Block Rate	Action Change Rate
Baseline	20.00%	4.00%	0.00%
Authentication Missing 20%	24.50%	4.83%	6.17%
Unseen Channel 10%	20.50%	3.83%	0.83%
Amount Increase 25%	20.50%	4.17%	0.83%
Distance Increase 50%	20.50%	4.17%	0.83%
Transaction Velocity +2	33.83%	7.33%	20.50%

The transaction velocity stress scenario produced the largest observed change in decisions.

Increasing transaction velocity by two resulted in approximately 20.50% of transaction decisions changing.

The review rate increased to 33.83%, exceeding the assumed 20% capacity limit.

The proposed block rate increased to 7.33%, exceeding the assumed 5% limit.

These results indicated that the system was sensitive to changes in transaction velocity.

Additional stress scenarios also resulted in review capacity violations.

Therefore, the fixed decision policy was not operationally stable under all tested conditions.

This finding highlighted the need for capacity-aware decision controls, monitoring systems, and human-approved overflow procedures.

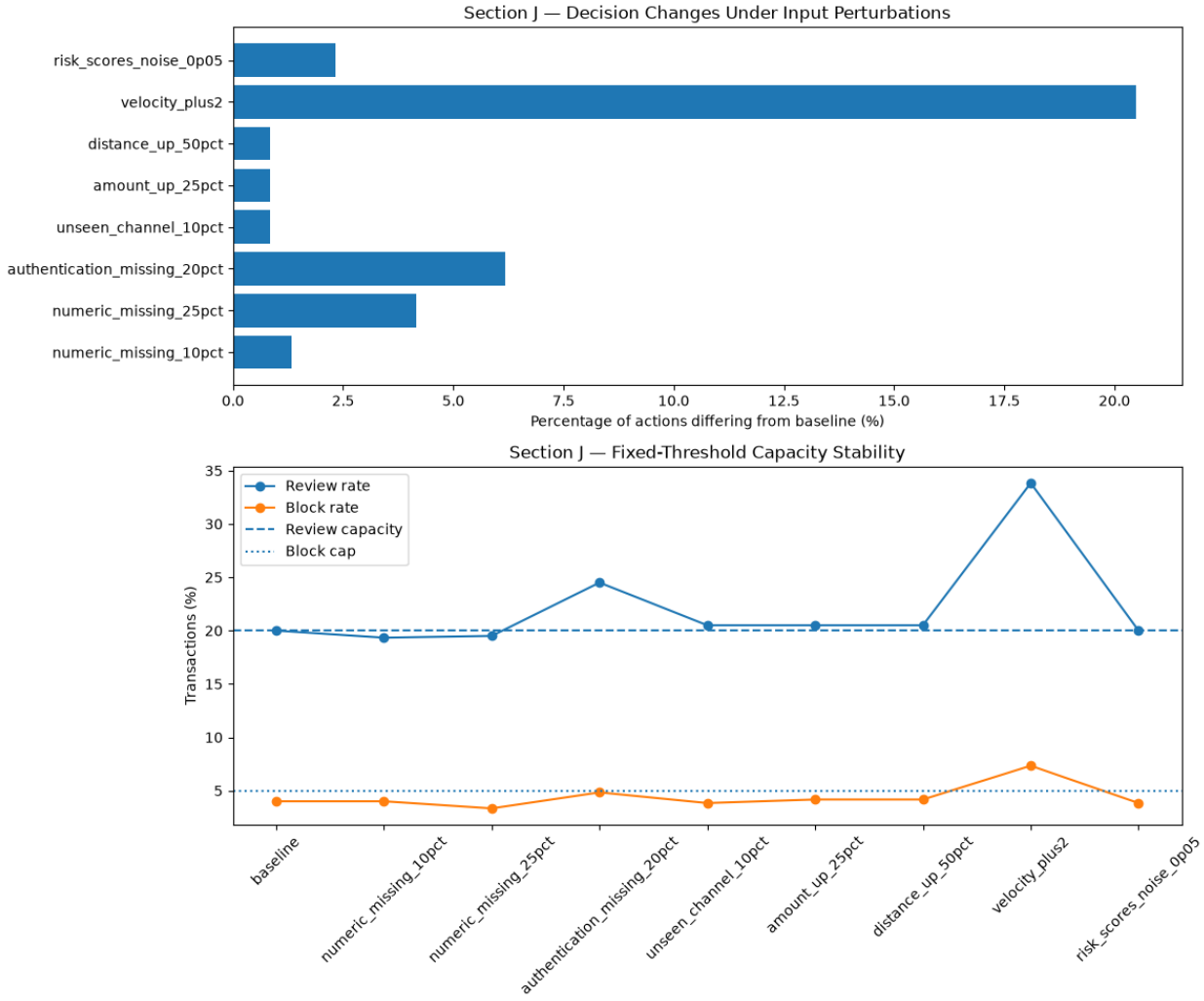


Figure 13.1 Decision Changes and Operational Capacity Stability Under Input Perturbations.

14. MODEL GOVERNANCE AND RISK MANAGEMENT

Model governance was incorporated to evaluate whether the framework was sufficiently reliable, transparent, and controlled for operational deployment.

A governance readiness assessment was conducted using ten evaluation gates.

At the governance checkpoint, only two gates passed.

The successful gates confirmed that the baseline validation batch remained within the assumed review and proposed block-rate limits.

Other gates remained unresolved due to issues involving operational stress capacity, prediction-time feature availability, temporal validation, business cost verification, fairness, appeals, monitoring, and independent evaluation.

The final holdout was subsequently evaluated once, resolving the outstanding execution of that assessment. However, this did not eliminate the remaining governance concerns.

14.1 Key Model Risks

Table 14.1 Major Model Risks and Mitigation Strategies

Risk	Severity	Recommended Mitigation
Fraudulent Transactions Approved	High	Improve detection and maintain layered controls
Legitimate Transactions Proposed for Block	High	Human approval and customer appeal procedures
Operational Capacity Overflow	High	Capacity monitoring and fallback routing
Prediction-Time Data Leakage	High	Verify feature availability before authorization
Limited Temporal Validation	High	Obtain timestamped and customer-level datasets
Unverified Business Costs	High	Collect actual fraud losses and investigation costs
Fairness and Customer Impact	High	Conduct subgroup impact assessments
Data Drift	Medium	Implement continuous monitoring
Limited Evaluation Sample	Medium	Conduct additional independent evaluations

14.2 Model Monitoring Recommendations

A future operational implementation should monitor transaction review rates, proposed blocking rates, missing authentication information, unseen categories, feature distribution changes, probability calibration, and delayed fraud outcomes.

Performance metrics should be evaluated using sufficiently mature fraud labels.

Monitoring thresholds should generate alerts when operational limits or predefined risk indicators are exceeded.

Model updates should require documented approval and independent validation.

14.3 Deployment Recommendation

The governance assessment resulted in the following decision:

RESEARCH ONLY — NOT APPROVED FOR LIVE AUTOMATED BLOCKING

This decision was supported by the substantial number of fraudulent transactions approved, legitimate transactions proposed for blocking, operational instability under stress, and unresolved data and governance requirements.

The framework should therefore remain an experimental decision-support prototype.

15. BUSINESS RECOMMENDATIONS

Based on the final evaluation, several recommendations are proposed.

15.1 Maintain Human Oversight

The current model should not independently approve high-impact blocking decisions.

Proposed blocking actions should require appropriate human verification until customer impact and detection reliability have been established.

Manual override and appeal procedures should be incorporated into future operational workflows.

15.2 Improve Fraud Detection Performance

The final testing results demonstrated limitations in fraud detection.

Future experiments should investigate additional transaction history, device behavior, customer-level patterns, and more representative datasets.

Model comparisons should consider precision-recall performance, calibration, business costs, and stability rather than accuracy alone.

15.3 Obtain More Representative Data

Future datasets should include complete timestamps and suitable customer identifiers to support chronological and customer-group validation.

This would help determine whether the model generalizes to new customers, emerging fraud patterns, and future transaction periods.

Additional independent datasets should be used before considering deployment.

15.4 Validate Business Cost Assumptions

The current decision policy depends on hypothetical financial costs and reviewer effectiveness.

Actual fraud losses, investigation expenses, false-blocking consequences, and reviewer outcomes should be measured.

These values should be incorporated into future cost-sensitive optimization experiments.

15.5 Introduce Operational Capacity Controls

Stress testing demonstrated that fixed thresholds could exceed investigation capacity.

Future implementations should include explicit review queue capacity controls, operational alerts, fallback procedures, and human-approved handling of excess cases.

Such controls should be evaluated under realistic workload conditions.

15.6 Establish Continuous Model Monitoring

A monitoring framework should be developed to track input data quality, fraud prevalence, prediction distributions, calibration, false positives, and fraud detection outcomes.

Model retraining should be considered when meaningful changes in data distributions or predictive performance are identified.

Any retraining or threshold adjustments should be assessed using new independent data.

16. PROJECT LIMITATIONS

Despite successfully implementing the major technical components, several limitations were identified.

First, the dataset contained only 3,000 transactions, limiting the reliability of some subgroup analyses and final evaluation estimates.

Second, complete transaction timestamps and customer identifiers were unavailable, preventing chronological and customer-level validation.

Third, several historical features required additional verification to ensure that their values would be available before transaction authorization.

Fourth, the business cost assumptions and manual review effectiveness values were hypothetical rather than institutionally verified.

Fifth, the final testing dataset contained only 73 fraudulent transactions, introducing uncertainty into model evaluation.

Sixth, the model achieved lower Average Precision on the final testing dataset than on the validation dataset.

Seventh, a significant proportion of fraudulent transactions remained in the approval category.

Eighth, the proposed blocking mechanism generated false blocking recommendations involving legitimate transactions.

Finally, operational stress testing demonstrated capacity violations under certain transaction behavior changes.

These limitations indicate that further development and evaluation are necessary before operational implementation.

17. FUTURE IMPROVEMENTS

Future development should focus on improving predictive reliability, operational effectiveness, and governance readiness.

Potential improvements include incorporating richer customer transaction histories, identifying behavioral anomalies using longitudinal data, and evaluating models on larger datasets.

Additional research could investigate alternative classification algorithms, time-aware validation, customer-level feature construction, and more comprehensive model calibration.

The decision optimization framework could also be extended to incorporate transaction-specific fraud losses and customer-dependent intervention costs.

Future explainability analysis should examine whether feature explanations remain stable across datasets, customer groups, and model versions.

Further operational research should evaluate review queue performance under changing workloads.

Finally, the framework should be assessed through independent external validation and monitored shadow testing before any production deployment is considered.

18. CONCLUSION

This project developed and evaluated an explainable, cost-aware Fraud Risk Intelligence Framework capable of generating calibrated fraud risk scores and supporting structured transaction decisions.

The implementation combined machine learning, feature engineering, class imbalance handling, probability calibration, financial cost optimization, explainability, fraud operations simulation, robustness testing, and model governance.

The final evaluation demonstrated that the selected model retained measurable fraud ranking capability, achieving a PR-AUC of 0.1932 and ROC-AUC of 0.6640.

The proposed decision framework also demonstrated potential economic value under the selected hypothetical business assumptions.

The final scenario-based evaluation estimated a cost reduction of approximately 31.66% compared with approving all transactions.

However, significant limitations remained.

The system approved 46 of the 73 fraudulent transactions and proposed blocking eight legitimate transactions.

Furthermore, stress testing demonstrated that review and blocking volumes could exceed operational limits under changing transaction conditions.

These findings indicate that the framework is not sufficiently reliable for live automated transaction blocking.

Nevertheless, the project successfully demonstrates how predictive modeling, probability calibration, financial decision analysis, operational simulation, and explainability can be integrated into a coherent fraud risk intelligence workflow.

The recommended next step is to strengthen data quality, obtain independent validation evidence, verify business costs, and establish appropriate human oversight before considering a monitored operational pilot.

The framework should therefore be regarded as a research prototype that provides a foundation for future development rather than a production-ready financial fraud detection system.

APPENDIX A — FINAL MODEL CONFIGURATION

Component	Final Configuration
Base Model	Logistic Regression
Class Imbalance Strategy	Unweighted
Probability Calibration	Sigmoid
Original Input Features	23
Dataset Split	60% / 20% / 20%
Final Testing Transactions	600
Review Threshold	0.1520464035
Block Threshold	0.2620026706
Review Capacity Limit	20%
Proposed Block Limit	5%
Deployment Status	Research Only

Model Version Verification

The frozen model was identified using a SHA-256 checksum to support reproducibility and prevent unintended model replacement during final evaluation.

Model SHA-256:

87217ee986fbed6763df7111294c051c94fb7934ea9ce3e185064c6841bce517

APPENDIX B — PROJECT EVIDENCE

The analysis and conclusions presented in this report were derived from the following project-generated evidence:

- Data Quality Investigation Report
- Advanced Exploratory Data Analysis Report
- Feature Engineering and Model Comparison Report
- Advanced Model Development Report
- Imbalanced Learning Evaluation
- Probability Calibration Evaluation
- Cost-Sensitive Learning Analysis
- Fraud Operations Simulation
- SHAP Explainability Analysis
- Robustness and Stress Testing Report
- Model Governance Report
- Final Holdout Evaluation Report

The project implementation also includes reusable Python modules, unit tests, a Jupyter Notebook, saved model artifacts, evaluation outputs, and model governance metadata.

The final holdout evaluation was performed using frozen model and policy settings.

No model retraining or threshold changes were performed based on the final testing results.